PDCBVC Codebase Map for RAG Prompts

Generated from `control_flow/inputs` only

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Scope

This guide is a prompt-area reference for RAG answers over the local COBOL sample. It is built only from files under `inputs/`: `inputs/cobol`, `inputs/copybooks`, `inputs/bms`, and `inputs/mapa`. Files under `temp/` are intentionally ignored.

High-Level System Map

Artifact	Meaning for RAG
`PDCBVC.CBL`	Main CICS COBOL program. It browses accounting/payment voice data and drives the screen flow for map `PDCBVC1`.
`PDCBVCM.BMS`	BMS mapset definition for the 24x80 terminal screen. It declares visible fields such as header, rows, selection input, messages, and PF-key hints.
`PDCBVCM.cpy`	Generated symbolic map copybook. Variables ending in `I` are input fields, ending in `O` are output fields, ending in `L` are field lengths, and ending in `A/F` are attribute/flag bytes.
`PDRTWA2.cpy`	Transaction Work Area layout. Most `TWCOB-` variables are persistent CICS navigation/context state shared across programs.
`PD1VOCI.cpy`	COMMAREA layout for external program `PD1VOCI`. In this codebase it is used to retrieve voice/accounting rows into table fields `PD1VOCI-TABVOX-*`.
`PD1FS00.cpy`	COMMAREA layout for external program `PD1FS00`. In this program it retrieves/validates session data.
`PDRUTI01.cpy`	Utility COMMAREA. Here it formats numeric amounts before display.
`PDRVC.cpy`	Voice/accounting record layout. Prefix `PDRVC-` means fields of an accounting voice record.
`PDRTIP01.cpy`	Table/voice type record layout. Prefix `T01-` means table entry metadata for voice classification and validity.
`result.txt`	MAPA structural facts: copybooks, SQL includes, DB2 table, program summary, and external calls. Use it as dependency evidence, not as path-level logic.

Prompt Rules

Use these rules when interpreting retrieved chunks:

- If a variable starts with `TWCOB-`, treat it as shared CICS transaction context from `PDRTWA2.cpy`. It usually carries user identity, current function, page state, selected voice, next program, and messages across screen transitions.
- If a variable starts with `PD1VOCI-`, treat it as the request/response area for `EXEC CICS LINK PROGRAM('PD1VOCI')`. It is the main source of browse rows displayed on the screen.
- If a variable starts with `PD1FS00-`, treat it as the request/response area for `EXEC CICS LINK PROGRAM('PD1FS00')`. In this program it is used for session/access information, including session year/month/type and session status.
- If a variable starts with `PDRUTI01-`, treat it as formatting/utility state. Here `PDRUTI01-FUNZIONE = '05'` is used before calling `PDOUTI01` to format amounts for display.
- If a variable starts with `PDRVC-`, interpret it as an accounting voice record field. Names containing `IMPI` refer to start/installation data; names containing `CESS` refer to cessation/end data; names containing `IMPORTO` are amounts.
- If a variable starts with `T01-`, interpret it as table metadata from `PDRTIP01.cpy`: voice code, applicability, description, validity period, and classification.
- If a variable starts with `PXCSEMAF-`, interpret it as semaphore/lock request state. The local call `CALL PXRSEMAF USING PXCSEMAF-AREA` checks whether insert/update operations are allowed.
- If a variable begins with `M` and appears in `PDCBVCM.cpy`, interpret it as a BMS map field. Suffix `I` means received input, `O` means output sent to the terminal, `L` means length, `A` or `F` means display attribute/flag.
- If the code writes `-1` to a `...L` field, it is positioning the CICS cursor on that field for the next send.
- If the code writes `ASKIP-DRK` to a `...A` field, it is making a map field dark/skipped, usually hiding unavailable PF-key hints.
- Treat `TWCOB-FASE` as a screen phase switch. In `PDCBVC`, phase `'1'` enters first browse rendering and phase `'2'` receives user input from the map.
- Treat `TWCOB-FUNZIONE` as the operation mode. In this program values are mapped to labels: `I`=IMPIANTO, `V`=VISUALIZZAZIONE, `C`=CESSAZIONE, `A`=AGGIORNAMENTO, `D`=MODIFICA DECORRENZA, `P`=MODIFICA PROVVEDIMENTO.
- Treat `TWCOB-VARCONT-NUMFUNZ` as the voice category selector. Values observed in the program map to: `1`=COMPETENZE E RITENUTE, `2`=COMPETENZE IMPONIBILI, `3`=COMPETENZE NON IMPONIBILI, `4`=RITENUTE PREVIDENZIALI, `5`=RITENUTE VARIE, `6`=VARIAZIONI MENSILI.
- Treat `WCTPAG`, `NPAGT`, `PD1VOCI-IND`, and `MAX-RIGHE` as pagination state. The screen shows up to `MAX-RIGHE = 15` rows using `MRIGAO(...)`.
- Treat `SCELTAI` as the user selection input field on row 22. The selected progressive number is matched against `WPROGR`; on match the program stores the selected voice and progressive number into `TWCOB-VARCONT-VOCE` and `TWCOB-VARCONT-PROGVOCE`.

- Treat M1MSGO as the primary message output field. Common messages include no records, invalid key/function, no selection, and end of data.
- Treat WABEND-CODE as local abnormal/error code routing. Values like BR00, LE10, FS00, UT01, and GET1 point to the failing local step or called service.

Control-Flow Rules

Paragraph / Area	Interpretation
BROWSE-FASE1	First screen phase. Initializes the map, validates update permission through semaphore logic when needed, prepares parameters, links to services, calculates pages, builds map rows, and sends the screen.
BROWSE-FASE2	Receive phase. Receives PDCBVC1, dispatches by EIBAID, handles ENTER/PF keys, pagination, selection, and navigation.
INIZ-PARAM	Builds the PD1VOCI request from TWCOB- context before linking to PD1VOCI.
LINK-PD1VOCI	Calls PD1VOCI. On return 'E', the program sets WABEND-CODE = 'LE10' and abends.
PREP-LINK-PD1FS00 / LINK-PD1FS00	Prepares and calls PD1FS00 for session/access data. Return '2' is accepted; return 'E' causes FS00 abend handling.
MUOVI-TESTATA	Fills screen header fields from TWCOB- and PD1FS00- fields, including date, system, person identity, session, and title text.
MUOVI-DATI / PREP-RIGA	Iterates PD1VOCI-TABVOX-* rows, formats dates and amounts, and moves row text into MRIGAO(WCTRIG).
READ-TAB-SEMAF	Checks semaphore state using PXRSEMAF; blocks insert/update when another process/state forbids it.
SEND-PDCBVC1 / RECEIVE-PDCBVC1	CICS terminal I/O for map PDCBVC1 in mapset PDCBVCM.
XCTL-LIV0..LIV5	Navigation exits. They reset TWA browse state, set TWCOB-FASE/FUNZIONE/NOME-PGM, then transfer to the target program flow.
RETURN-MAIN / RETURN-CICS	Save TWA if needed, issue syncpoint, and return to CICS/transaction PRED.

External Dependency Rules

MAPA reports these structural dependencies. Prefer these names when building RAG context:

- PD1VOCI: CICS LINK by literal. Main data retrieval service for accounting voice rows.
- PD1FS00: CICS LINK by literal. Session/access service.
- PDOUTI01: CICS LINK by literal. Utility/formatting service.
- PXRSEMAF: dynamic CALL by identifier. Semaphore/lock check.
- PDPRED: CICS XCTL by literal. Main transfer target.

- DUAL: DB2 table used only for SELECT SYSDATE INTO :HSQL-DATE FROM DUAL.

BMS Field Rules

Field Pattern	Meaning
PDCBVC1I	Full input map area for PDCBVC1. It is cleared before first send with MOVE LOW-VALUE TO PDCBVC1I.
PDCBVC1O	Output overlay for the same map area. Values moved to ...O fields are displayed on screen.
M1SYSO	System identifier displayed in the header. Filled from TWCOB-ID-SISTEMA.
M1DATAO	Display date. Filled after CICS ASKTIME/FORMATTIME.
MOAMBO	CICS environment/name field. Filled from TWCOB-NOME-CICS.
MMATRO, MCOGO, MNOMO	Person identity fields: matricola/codice, surname, and name from TWCOB-.
MSESSIO, MDSESSO	Session code and session status description. Status flag 1=Aperta, 2=Chiusa, 3=Liquidata.
MNPAGO	Current page/total page display, built from WCTPAG and NPAGT.
MDESCO	Screen title/description, built from operation and category labels.
MRIGAO(1..15)	Repeated output rows. Each row is assembled in RIGA-MAPPA and moved into MRIGAO(WCTRIG).
SCELTAI/SCELTAO	Two-character user selection input/output. SCELTAL = -1 positions cursor here.
M1MSGO	Message line. Used for validation errors, no data, and end-of-data messages.
M1PF7A, M1PF8A	PF-key display attributes. The program darkens them when previous/next page is unavailable.

Data Movement Rules

- The main browse data path is: TWCOB-* context → PD1VOCI-* request → PD1VOCI-TABVOX-* response → local RIGA-MAPPA → MRIGAO(...) → terminal map.
- The session/header path is: TWCOB-ID-SISTEMA → PD1FS00-SESS-* → MSESSIO/MDSESSO/M1SYSO.
- The amount formatting path is: raw amount from PD1VOCI-TABVOX-RATA(...) → PDRUTI01-F05-* via PDOUTI01 → IMPORTO-RATA → RIGA-MAPPA.
- Selection path is: user enters SCELTAI → program scans row MRIGAO(WCTRIG) → matching WPROGR stores TWCOB-VARCONT-VOCE and TWCOB-VARCONT-PROGVOCE for downstream programs.
- Navigation path is: PF key or selected action → XCTL-LIV* paragraph → TWCOB-NOME-PGM set from PROG-LIV* constants → CICS transfer/return flow.

RAG Answering Instructions

When answering questions about this codebase:

1. Always identify whether a variable comes from main working storage, a copybook COMMAREA, the TWA, or the BMS map.
2. For UI questions, join evidence from `PDCBVC.CBL`, `PDCBVCM.BMS`, and `PDCBVCM.cpy`. BMS field names alone are not enough; use code moves to explain runtime values.
3. For business behavior, prefer paragraph names and data movement paths over isolated lines. Example: explain row display through `MUOVI-DATI` and `PREP-RIGA`.
4. For dependency questions, use MAPA `result.txt` as the authoritative list of copybooks, calls, SQL includes, and DB2 table references.
5. Do not infer runtime certainty from MAPA alone. MAPA says what can be called statically; paragraph logic and conditions decide when it is called.
6. When a field has a suffix convention, explain it explicitly: `I/O/L/A/F` for BMS, `RETURN/FUNZIONE` for service COMMAREAs, `FASE/FUNZIONE/VARCONT` for TWA state.
7. If a question asks "what does this variable mean", first map by prefix, then inspect its surrounding moves and conditions.